



VIT[®]

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Agile Development Process and Devops Lab

ASSESSMENT-3

MAVEN

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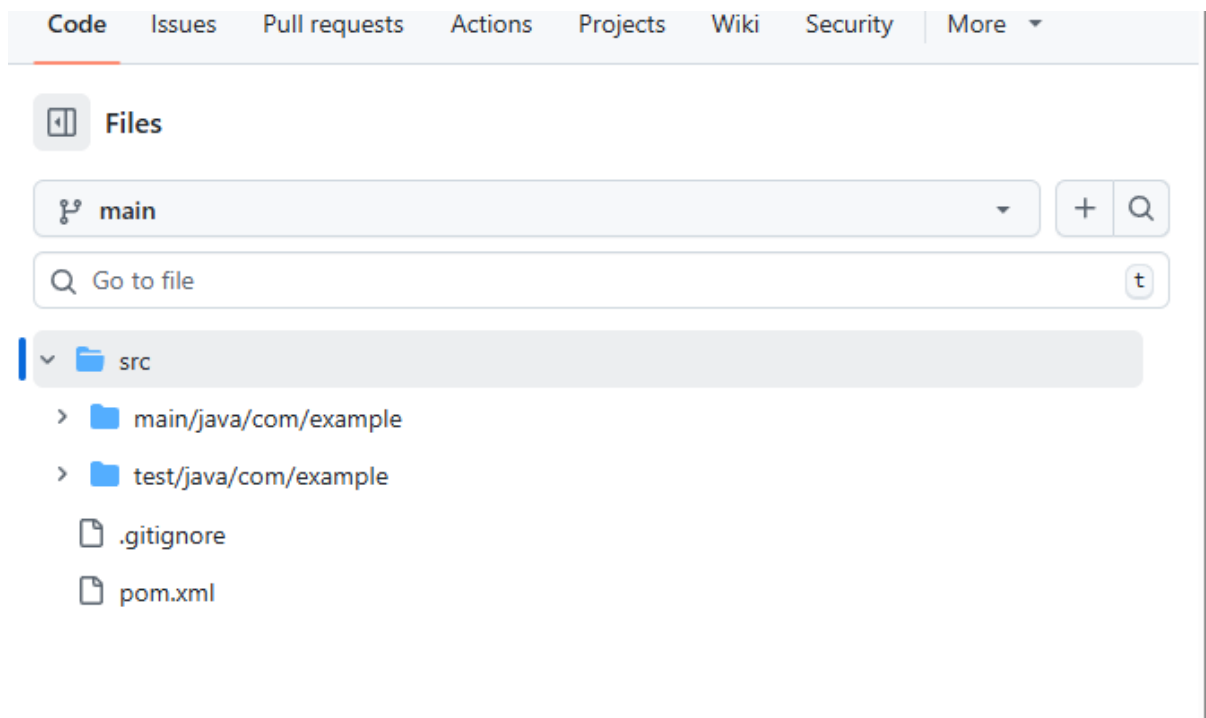
1. Create a Maven Project in Git

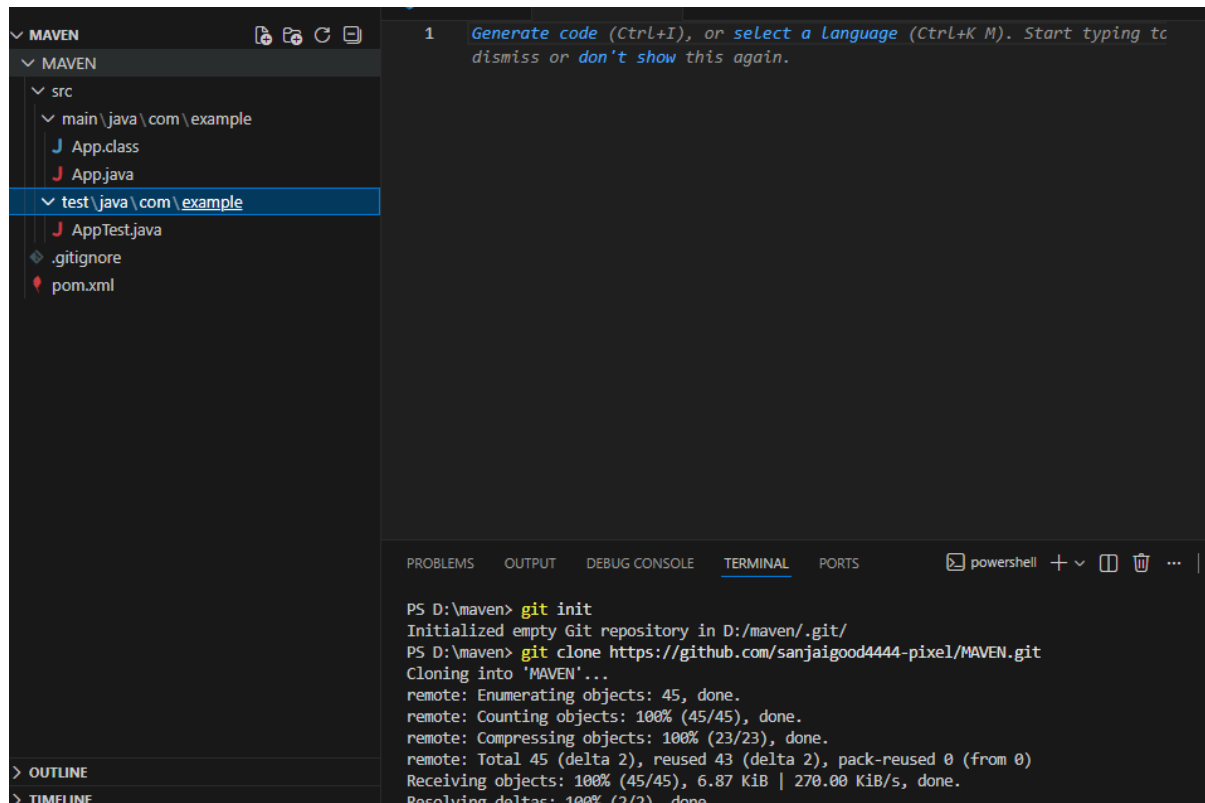
- Create a new Git repository
- Create a simple Maven project structure manually, including the following files and folders:

o pom.xml

o src/main/java/com/example/App.java

o src/test/java/com/example/AppTest.java





2. Implement the Java Program

- Write a simple Java program in App.java under your newly created Git repository for any one of the following scenarios:

o Student Grade Calculator

App.java

```
MAVEN > src > main > java > com > example > J App.java > Language Support for Java(TM) by Red Hat > App > subtract(int, int)
1 package com.example;
2
3 public class App {
4
5     public int add(int a, int b) {
6         return a + b;
7     }
8
9     public int subtract(int a, int b) {
10        return a - b;
11    }
12
13    public int multiply(int a, int b) {
14        return a * b;
15    }
16
17    public int divide(int a, int b) {
18        if (b == 0) {
19            throw new ArithmeticException("Cannot divide by zero");
20        }
21        return a / b;
22    }
23
24    Run main | Debug main | Run | Debug
25    public static void main(String[] args) {
26        App calc = new App();
27    }
```

- Write a corresponding unit test in AppTest.java.

```
MAVEN > src > test > java > com > example > J AppTest.java > Language Support for Java(TM) by Red Hat > AppTest
1 package com.example;
2
3 import static org.junit.Assert.assertEquals;
4 import org.junit.Test;
5
6 public class AppTest {
7
8     App calc = new App();
9
10    @Test
11    public void testAddition() {
12        assertEquals(15, calc.add(a: 10, b: 5));
13    }
14
15    @Test
16    public void testSubtraction() {
17        assertEquals(5, calc.subtract(a: 10, b: 5));
18    }
19
20    @Test
21    public void testMultiplication() {
22        assertEquals(50, calc.multiply(a: 10, b: 5));
23    }
24
25    @Test
26    public void testDivision() {
27        assertEquals(2, calc.divide(a: 10, b: 5));
28    }
29 }
```

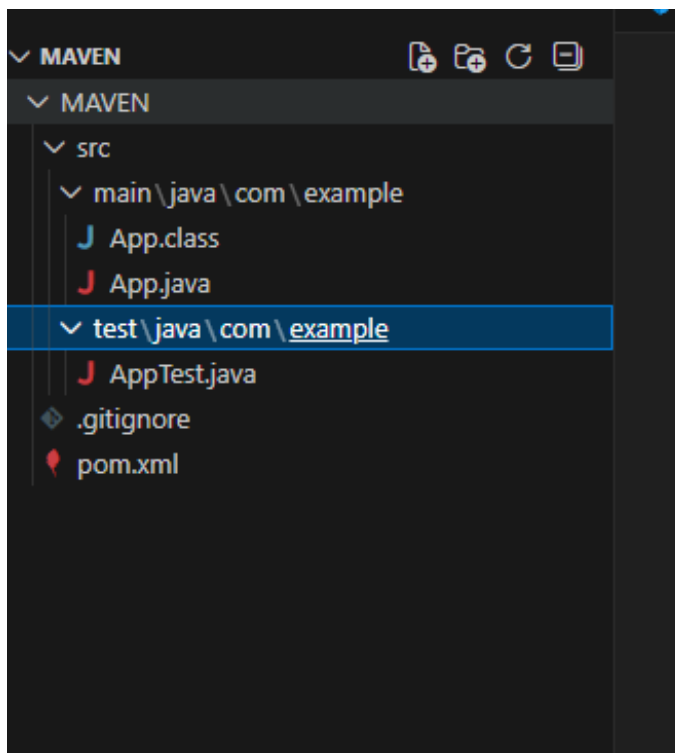
3. Push and Clone the Repository

- Clone the repository using Visual Studio Code.

```
PS D:\maven> cd maven
PS D:\maven\maven> git pull
Already up to date.
PS D:\maven\maven> git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS D:\maven\maven> █
```

- Verify the project structure.



4. Build the Maven Project

- Use Maven commands in Visual Studio Code terminal to:

- o Compile the project
- o Run tests
- o Build the project (mvn clean install)

Compile the project:

```
PS D:\maven\maven> mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:simple-maven-app >-----
[INFO] Building simple-maven-app 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ simple-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory D:\maven\MAVEN\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ simple-maven-app ---
[INFO] Recompiling the module because of added or removed source files.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.822 s
[INFO] Finished at: 2026-03-04T10:15:20+05:30
[INFO] -----
```

Run tests:

```
[INFO] -----
PS D:\maven\maven> mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:simple-maven-app >-----
[INFO] Building simple-maven-app 1.0-SNAPSHOT
[INFO]    from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ simple-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory D:\maven\MAVEN\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ simple-maven-app ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ simple-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory D:\maven\MAVEN\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ simple-maven-app ---
[INFO] Recompiling the module because of added or removed source files.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\test-classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ simple-maven-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AppTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.177 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time:  6.432 s
[INFO] Finished at: 2026-03-04T10:15:46+05:30
[INFO] -----
```

Build the project (mvn clean install):

```
PS D:\maven\maven> mvn clean install
[INFO] Scanning for projects...
[INFO] -----< com.example:simple-maven-app >-----
[INFO] Building simple-maven-app 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-jar-plugin/3.4.1/maven-jar-plugin-3.4.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-jar-plugin/3.4.1/maven-jar-plugin-3.4.1.pom (7.8 kB at 7.2 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/42/maven-plugins-42.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/42/maven-plugins-42.pom (7.7 kB at 94 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/42/maven-parent-42.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/42/maven-parent-42.pom (50 kB at 639 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/32/apache-32.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/32/apache-32.pom (24 kB at 341 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/3.1.2/maven-install-plugin-3.1.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/3.1.2/maven-install-plugin-3.1.2.pom (7.7 kB at 94 kB/s)
```

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AppTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.180 s -- in com.example.AppTest
[INFO] Results:
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO] --- jar:3.4.1:jar (default-jar) @ simple-maven-app ---
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/file-management/3.1.0/file-management-3.1.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/file-management/3.1.0/file-management-3.1.0.pom (4.5 kB at 51 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-components/36/maven-shared-components-36.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-components/36/maven-shared-components-36.pom (4.9 kB at 62 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/36/maven-parent-36.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/36/maven-parent-36.pom (45 kB at 640 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/26/apache-26.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/26/apache-26.pom (21 kB at 316 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus-utils-4.0.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus-utils-4.0.1.pom (7.8 kB at 93 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.16.1/commons-io-2.16.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.16.1/commons-io-2.16.1.pom (20 kB at 302 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-parent/69/commons-parent-69.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-parent/69/commons-parent-69.pom (77 kB at 48 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-archiver/3.6.2/maven-archiver-3.6.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-archiver/3.6.2/maven-archiver-3.6.2.pom (4.4 kB at 52 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-components/41/maven-shared-components-41.pom
```

```
[INFO] BUILD SUCCESS
[INFO] Total time: 16.584 s
[INFO] Finished at: 2026-03-04T10:16:46+05:30
[INFO] PS D:\maven\maven>
```


5. Create and Run Jenkins Pipeline

- Create a new Pipeline project in Jenkins.

New Item

Enter an item name

maven1

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.



Multibranch Pipeline

Creates a set of Pipeline projects according to detected branches in one SCM repository.

OK

- Create a pipeline script

Define your pipeline using Groovy directly or pull it from source control

Definition

Pipeline script

Script ?

```

1 pipeline {
2   agent any
3
4   tools {
5     maven 'Maven'
6     jdk 'JDK'
7   }
8
9   stages {
10
11     stage('Clone Repository') {
12       steps {
13         git 'https://github.com/sanjaigood4444-pixel/MAVEN.git'
14       }
15     }
16   }
17 }

```

try sample Pipeline...

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

- Execute the pipeline and verify successful build completion.

Jenkins / maven1

Status

Changes

Build Now

Configure

Delete Pipeline

Full Stage View

Favorite

Open Blue Ocean

Stages

✓ maven1

Stage View

Average stage times:
(full run time: ~1min 20s)

	Clone Repository	Compile	Test	Build
5s	5s	6s	12s	
Mar 04 11:10 No Changes	5s	16s	19s	36s